The Monk Skin Tone Scale (MST)

Dr. Ellis P. Monk, Jr
Harvard University

Human skin is the most visible aspect of human phenotype (Jablonski, 2004). Among human beings there is a great variety of different skin colors, which are roughly correlated with the distribution of UV radiation around the world. Specifically, given that human skin acts as a shield against the sun to protect against solar UV radiation, darker-skin tones are generally found among those populations with origins that trace back to the areas on Earth with the highest amount of exposure to UV from the sun (Jablonski, 2021). Unfortunately, despite there being no biological or genetic connection between skin color and human ability, there has been a long history of ostensibly scientific classifications of human beings, which have used skin color as the 'cardinal characteristic distinguishing people' (Jablonski, 2021; Monk Jr, 2022). From Linnaeus's Systema Naturae (1735) to von Luschan's chromatic scale (1897; also known as the von Luschan skin color scale), skin color has been associated with the classification of human beings into various races, often with the implication of a hierarchical ranking.

Given the centuries-long history of ethnoracial oppression, discrimination, and inequality, social scientists have also sought to measure skin color to assess the extent to which differences in human skin color within and across various ethnoracial populations are associated with a whole range of important life outcomes from education to earnings to health and more (Monk Jr, 2021). The study of skin tone stratification and colorism, which is discrimination on the basis of skin tone, has a long history in the social sciences, especially sociology and economics(Monk Jr, 2021). This research consistently shows that even within the same ethnoracial category, there is often significant inequality in education, earnings, health, and much more. In fact, on some outcomes there is more inequality within the same ethnoracial category associated with skin color differences than there is between ethnoracial categories as a whole (Monk Jr, 2021). Moreover, it is important to note that skin tone stratification and colorism are global phenomena – research finds strong evidence of skin tone stratification in countries such as Brazil and India (Monk Jr, 2016, 2021).

Until very recently, the measurement of skin color for machine learning and artificial intelligence has drawn heavily from dermatology and not the social sciences. Specifically, for the purposes of data annotation, the use of the Fitzpatrick Scale (FST) has been nearly ubiquitous. The Fitzpatrick Scale is a questionnaire designed by Dr. Thomas B. Fitzpatrick (Harvard Medical School) in 1975 to determine the appropriate dose of oral methoxsalen for treating psoriasis using photochemotherapy among White Americans (Fitzpatrick, 1988). The FST is commonly used, to this very day, by dermatologists around the world, not only for its original purpose, but also the classification and measurement of different human skin colors. In recent decades, however, the FST has come under heavy criticism; and, in many ways the growing chorus of criticism the Fitzpatrick Scale has received in the realm of dermatology (Ware et al., 2020), mirrors criticism in machine learning and artificial intelligence. Data set construction, data annotation, and, ultimately, the algorithms that flow from these processes in facial recognition software, self-driving cars recognition of their environments, automatic image classification in photos and videos (e.g., skin tone classification, gender classification, etc.), and more, are all vulnerable to biases when the standard of measurement for skin tone is not as inclusive as it could or should be. For example, lack of comprehensive and diverse training data sets (in terms of race and skin tone) is one leading explanation for facial recognition inaccuracies (e.g., spurious matches) and inaccuracies in automated, machine-learning driven image and video classifiers. In short, similar issues plague both pulse oximetry and machine learning and artificial intelligence (e.g., light-sensing and computer vision).

Again, it is worth noting that the FST was not designed to classify a diverse range of skin tones, but instead as a tool to use when considering how skin reacts to phototherapy (UV) among Whites (Ware et al., 2020). Only a decade after its launch was the scale expanded from 4 skin tones to 6 skin tones. It should not be surprising, then, that studies find, for example, that the Fitzpatrick Scale "excludes the majority of Blacks and yields data that overestimate Black population prevalence of type IV skin" (Pichon et al., 2010). Similarly, Sommers et al. (2019) summarizes this line of work criticizing the Fitzpatrick Scale by noting that "[Fitzpatrick Scale] provides a restricted range of options for people with darker skin tones that do not capture variations in their skin color." It is also worth noting that some studies find that the Fitzpatrick Scale performs poorly, even when used as intended (see above), especially on ethnoracial minorities Eilers et al. (2013); He et al. (2014). Ironically, there is even reason to believe that the FST is simultaneously too

dark for most lighter-skinned humans [FST I-II] and not dark enough for most darker-skinned humans [FST IV-VI].

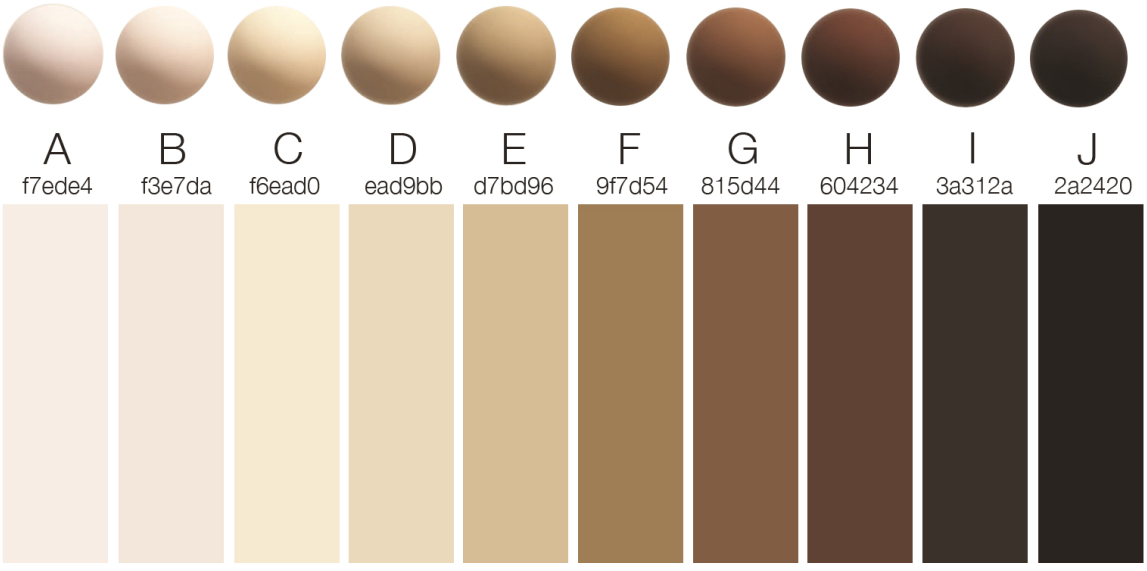
The Monk Skin Tone Scale (MST) is a measure of skin color from the social sciences, which was explicitly designed to capture variation in human skin color in the Americas (and potentially beyond) for the purposes of examining skin tone stratification and colorism. The main way the Monk Skin Tone Scale (MST) mitigates existing biases found in other skin color scales is by including a wider range of carefully selected skin tones to better represent the dynamic range of skin tones we see in the Americas (and beyond). As explained above, the Monk Skin Tone Scale was specifically designed for skin tone classification in a diverse population; and evidence from validation research already shows that, according to a nationally-representative sample of respondents from the U.S., the Monk Scale is indeed perceived as significantly more inclusive than the Fitzpatrick Scale(Heldreth et al., 2022). These results, in addition to internal testing at Google, which showed greater classificatory efficiency and accuracy using the Monk Scale compared to their previous skin tone models, which were based on the FST, were the basis of its adoption by Google.

Similarly, the Monk Scale has been adopted as an alternative to the Fitzpatrick Scale for skin tone classification by Facebook/Meta in their Casual Conversations data Hazirbas et al. (2022). Casual Conversations is a comprehensive data set designed to help ensure robust and fair AI systems that are un-biased. The Monk Skin Tone Scale (MST), then, is a promising alternative to both the Fitzpatrick Scale, the von Luschan Scale, and other alternative scales in the social sciences (e.g., Massey-Martin). Suggestions about how to use the scale are available below.

Monk Scale

How to use: This scale is intended to be used to classify human skin color. **Human skin color is immensely variable and complex, but scales, such as this one, are often used as a pragmatic and cost-effective means of approximately measuring common differences in skin color** among human beings. **For some use-cases, printing the scale may be necessary.** Print this page using color acccurate printing. Once printed, **find the most appropriate match between an orb and/or swatch and the desired site** (e.g., face, back of hand, etc). Holding the card near the site may help accomplish this. For instance, depending on your use-case, consider using a hole punch to punch a hole in the center of each orb and/or the center of each rectangle. Hold the printed card over the skin at the site you wish to describe. Move the card to find the color that matched most closely and note the category letter (e.g. 'E'). **The orbs or squares may be used and you should note in your methods which you have chosen.** As you may notice, the orbs have more shades per category, which better resemble human skin and may aid in finding the closest match.

Suggested citation: Monk, Ellis. 2019. “Monk Skin Tone Scale.” Google https://skintone.google



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